

Lab Assignment 6.1

AIAC

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Task 1:

```
1 # Task 1: Print all even numbers between 1 and N
2
3 N = int(input("Enter a number: "))
4
5 print("Even numbers between 1 and", N, "are:")
6 for i in range(1, N + 1):
7     if i % 2 == 0:
8         print(i)
9
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
G SHASHI KUMAR@LAPTOP-42G08GQ2 MINGW32 ~
$ "C:/Users/G SHASHI KUMAR/AppData/Local/Microsoft/WindowsApps/python3.11.exe" "c:/Users/G SHASHI KUMAR/OneDrive/Desktop/AIAC/t1-4.4.py"
Enter a number: 5
Even numbers between 1 and 5 are:
2
4
```

Task 2:

Generate Python code that counts how many numbers are even and odd in a list.
Use a loop and if-else condition.
Print both counts.

 Add Context...

```
2 # Task 2: Count even and odd numbers in a list
3
4 numbers = [10, 15, 20, 25, 30, 33]
5
6 even_count = 0
7 odd_count = 0
8
9 for num in numbers:
10     if num % 2 == 0:
11         even_count += 1
12     else:
13         odd_count += 1
14
15 print("Even numbers count:", even_count)
16 print("Odd numbers count:", odd_count)
17
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
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$ "C:/Users/G SHASHI KUMAR/AppData/Local/Microsoft/WindowsApps/python3.11.exe" "c:/Users/G SHASHI KUMAR/OneDrive/Desktop/AIAC/t1-4.4.py"
Even numbers count: 3
Odd numbers count: 3
```

Task 3:

Validate age using conditional statements.
Validate email by checking '@' and '.'.
Print whether the input is valid or invalid.

 Add Context...

```
2
3 class User:
4     def __init__(self, age, email):
5         self.age = age
6         self.email = email
7
8     def validate_age(self):
9         if self.age > 0:
10             return True
11         else:
12             return False
13
14     def validate_email(self):
15         if "@" in self.email and "." in self.email:
16             return True
17         else:
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
G SHASHI KUMAR@LAPTOP-42G08GQ2 MINGW32 ~
$ "C:/Users/G SHASHI KUMAR/AppData/Local/Microsoft/WindowsApps/python3.11.exe" "c:/Users/G SHASHI KUMAR/OneDrive/Desktop/AIAC/t1-4.4.py"
User details are valid
User details are invalid
```

Task 4:

Create a Python class Student with attributes name, roll_number, and marks.
Add methods to calculate total marks and average marks.
Display student details.

Add Context...

```
2
3 class Student:
4     def __init__(self, name, roll_number, marks):
5         self.name = name
6         self.roll_number = roll_number
7         self.marks = marks
8
9     def total_marks(self):
10        | return sum(self.marks)
11
12    def average_marks(self):
13        | return self.total_marks() / len(self.marks)
14
15    def display_details(self):
16        | print("Name:", self.name)
17        | print("Roll Number:", self.roll_number)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
G SHASHI KUMAR@LAPTOP-42G08GQ2 MINGW32 ~
$ "C:/Users/G SHASHI KUMAR/AppData/Local/Microsoft/WindowsApps/python3.11.exe" "c:/Users/G SHASHI KUMAR/OneDrive/Desktop/AIAC/t1-4.4.py"
Name: Rahul
Roll Number: 101
Total Marks: 245
Average Marks: 81.66666666666667
```

Task 5:

Generate a complete Python program for a simple bank account system.
Use class, loops, and conditional statements.
Include deposit, withdraw, and balance check options.

Add Context...

```
2
3 class BankAccount:
4     def __init__(self, name, balance=0):
5         self.name = name
6         self.balance = balance
7
8     def deposit(self, amount):
9         if amount > 0:
10            self.balance += amount
11            print("Amount deposited successfully")
12        else:
13            print("Invalid deposit amount")
14
15    def withdraw(self, amount):
16        if amount <= self.balance:
17            self.balance -= amount
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
G SHASHI KUMAR@LAPTOP-42G08GQ2 MINGW32 ~
$ "C:/Users/G SHASHI KUMAR/AppData/Local/Microsoft/WindowsApps/python3.11.exe" "c:/Users/G SHASHI KUMAR/OneDrive/Desktop/AIAC/t1-4.4.py"

1. Deposit
2. Withdraw
3. Check Balance
4. Exit
Enter your choice: 1
Enter amount to deposit: 500
Amount deposited successfully
```